FAKE NEWS DETECTION SYSTEM

Machine Learning Project

PROJECT OVERVIEW

- Developed a Machine Learning model to identify fake and real news articles.
- Used Natural Language Processing (NLP) techniques for text preprocessing.
- Built a user-friendly web application using Streamlit.
- Helps users verify whether a news article is likely to be Fake or Real.

PROBLEM STATEMENT

- With the rapid growth of online news and social media, fake news spreads quickly and can mislead people.
- This project aims to automatically classify news articles as **Fake** or **Real** using Machine Learning.

TECHNOLOGIES USED

- Python
- Pandas
- NumPy
- Scikit-learn
- NLP (Natural Language Processing)
- TF-IDF Vectorizer
- Logistic Regression
- Streamlit
- Pickle

DATASET

- **Dataset Used:**

  Fake.csv

  True.csv

- **Dataset Contains:**

  News Title

  News Text

  Subject

  Date

  Label (Fake / Real)

PROJECT WORKFLOW

1. Data Collection
2. Data Cleaning
3. Text Preprocessing
4. TF-IDF Vectorization
5. Model Training (Logistic Regression)
6. Model Evaluation
7. Streamlit Deployment

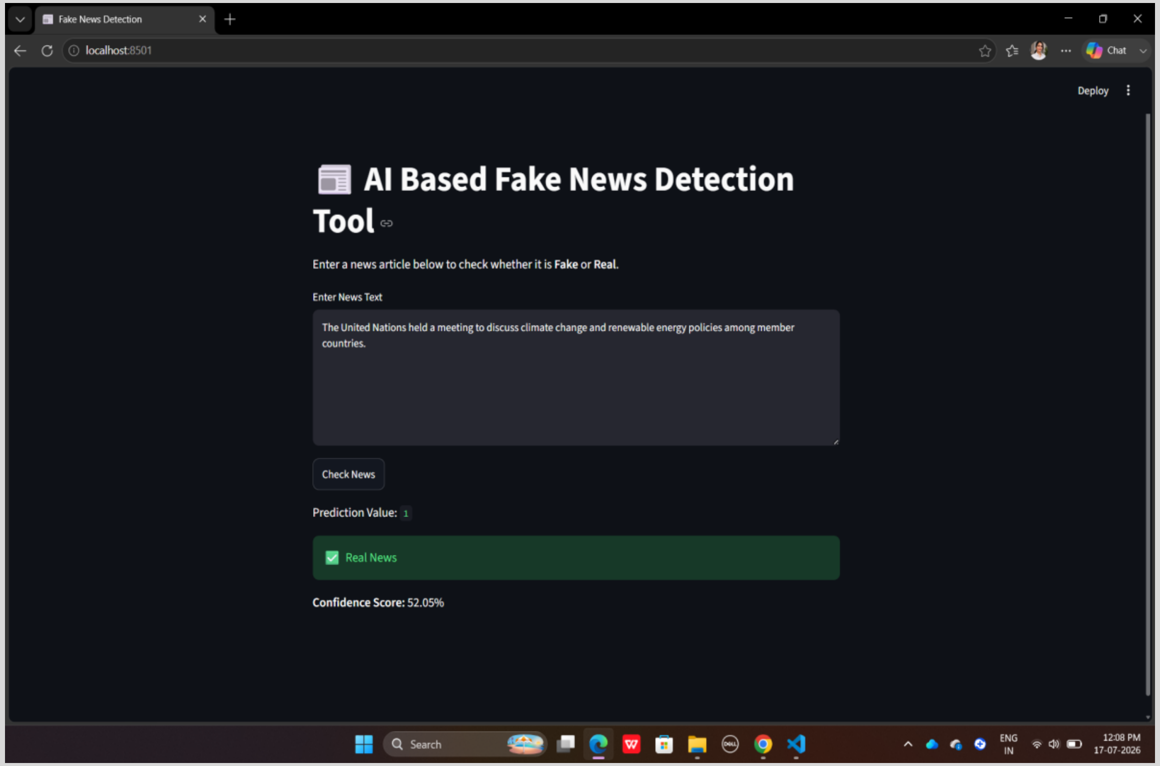
FEATURES

- User enters a news article.
- Detects whether the news is Fake or Real.
- Machine Learning-based prediction.
- Simple and interactive Streamlit interface.
- Fast prediction results.

RESULTS

Project Output

- Successfully classifies news articles.
- High prediction accuracy on the test dataset.
- Interactive Streamlit application.
- Trained model saved using Pickle.



CONCLUSION

- Successfully developed a Fake News Detection System using Machine Learning.
- Improved understanding of NLP techniques and text classification.
- Learned data preprocessing, feature extraction, model training, and deployment using Streamlit.

THANK YOU!